

Summary:

- Data Engineer with 3+ years of experience delivering scalable data solutions in healthcare and medical devices industries.
- Skilled in designing end-to-end ETL pipelines, transforming raw data into actionable insights, and supporting analytics & ML initiatives.
- Proficient in Python, SQL, Spark/PySpark, Airflow, Kafka, and Snowflake, with hands-on experience across AWS, Azure, and GCP.
- Strong background in data warehousing, workflow orchestration, and big data processing, ensuring accuracy, scalability, and compliance with healthcare standards.
- Adept at collaborating with cross-functional teams to build dashboards and data products that drive operational efficiency and strategic decisions.

Skills:

Programming & Scripting: Python, SQL, PySpark, Shell Scripting

Databases & Data Storage: MySQL, PostgreSQL, Snowflake, MongoDB, Hadoop, HDFS, Hive

Data Engineering & ETL / Pipeline Orchestration: Apache Spark, Airflow, Kafka, dbt

Cloud Platforms & Data Warehouses: AWS (S3, EC2, Redshift), Azure (SQL, Data Factory, Synapse), GCP (BigQuery) Tools, BI & Version Control: Tableau, Power BI, Git, REST APIs, Data Modeling (star & snowflake schemas)

Experience:

UnitedHealth Group

Aug 2024 – Present

Data Engineer

Responsibilities

- Designed and built robust ETL pipelines using PySpark to ingest, process, and stage healthcare records per month from multiple sources (claims, patient data, EHRs), improving data freshness for analytics by ~40%.
- Implemented data quality monitoring (missing values, schema drift) and alerting, reducing upstream data errors by ~30%.
- Collaborated with data science teams to productionize ML models (e.g. patient risk scoring, cost prediction) using feature pipelines; managed feature engineering, deployment, and ongoing monitoring.
- Migrated parts of the legacy on-prem data warehouse to Snowflake / cloud based warehousing, optimizing storage and query performance; reduced query run times on key dashboards by ~50-60%.
- Partnered with stakeholders (clinical operations, finance) to define KPIs, dashboard requirements; built dashboards in Tableau / Power BI that enabled non-technical teams to monitor trends and make strategic decisions.

Meril

Jan 2022 – June 2023

Junior Data Engineer

Responsibilities

- Developed ETL workflows to consolidate device manufacturing, supply chain, and quality control data from multiple source systems (SQL, flat files) into a central data lake using Spark & Hive, improving reporting lead time from days to hours.
- Wrote SQL queries and stored procedures for data extraction and transformations; cleaned and normalized data to ensure consistency across product lines.
- Built dashboards (Tableau / Power BI) for operations teams to monitor key manufacturing metrics (yield, downtime, defect rates), enabling identification of bottlenecks that improved throughput by ~15%.
- Assisted in maintaining on-prem Hadoop cluster and facilitated data ingest from IoT / sensor data to support product performance analytics.

Additional Qualifications & Highlights

- Experience working in highly regulated / compliance sensitive environments (healthcare, medical device), including handling PHI / PII, ensuring data security and privacy.
- Strong communication and collaboration skills; have worked cross-functionally with clinical, finance, R&D, operations.
- Familiar with data modeling, using star schemas / snowflake schemas; designing both batch and incremental data pipelines.
- Agile development experience, version control (Git), CI/CD principles.

Education:

- Master's in Data Science, University of Maryland, Baltimore County (UMBC) Baltimore, MD